



# CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Main)

UNIT TEST

24-07-2022

**JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & LIVE-I(A)]**

Time : 3 Hours

**PAPER-2**

Maximum Marks : 300

**READ THE INSTRUCTIONS CAREFULLY**
**Important Instructions :**

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The Test Booklet consists of **90** questions.
4. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **Two sections**.
  - (i) Section-I contains 20 multiple choice questions with **only one correct** option.  
**Marking scheme** : +4 for correct answer, 0 if not attempted and -1 in all other cases.
  - (ii) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions.  
 First 5 attempted questions will be considered for marking.  
**Marking scheme** : +4 for correct answer, 0 if not attempted and -1 in all other cases.
5. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** and **Side-2** of the Answer Sheet. **Use of pencil is strictly prohibited.**
6. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
7. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
8. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
9. **Do not fold or make any stray marks on the Answer Sheet.**
10. Take  $g = 10 \text{ m/s}^2$  unless otherwise stated.

DO NOT BREAK THESE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Name of the Candidate (in Capitals)\_\_\_\_\_

Form Number : in figures\_\_\_\_\_

: in words\_\_\_\_\_

Centre of Examination (in Capitals) : \_\_\_\_\_

Candidate's Signature : \_\_\_\_\_ Invigilator's Signature : \_\_\_\_\_

**Your Target is to secure Good Rank in JEE(Main) 2023**
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SECTION-I : (Maximum Marks: 80)

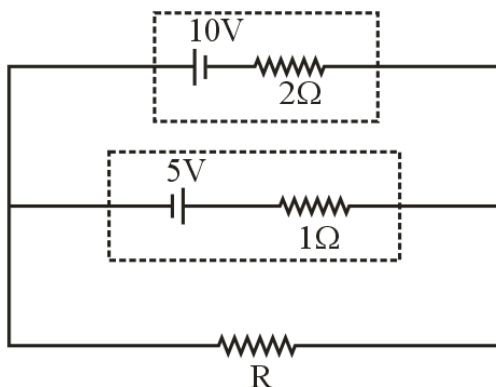
This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

**Full Marks** : +4 If correct answer is selected.

**Zero Marks** : 0 If none of the option is selected.

**Negative Marks** : -1 If wrong option is selected.

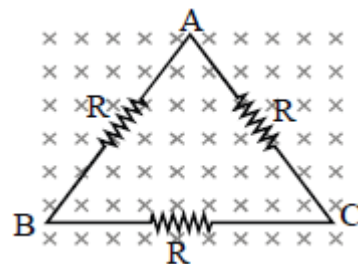
1. A circuit diagram is shown in figure.



Which of the following option is incorrect.

- (A) Current flowing in resistance R is zero.  
 (B) Terminal voltage of 10 V battery is zero.  
 (C) Terminal voltage of 5 V battery is zero.  
 (D) Current flowing through battery of 10 V is zero.
2. A charged particle of mass 1 kg and charge 1 C moves with velocity  $\vec{v} = 6\hat{i} + 8\hat{j}$  m/s in a magnetic field of  $\vec{B} = 2\hat{k}$  T. Then :
- (A) the path of particle may be  $y^2 + z^2 = 25$   
 (B) the path of particle may be  $x^2 + z^2 = 25$   
 (C) the path of particle may be  $x^2 + y^2 = 25$   
 (D) time period of particle will be 2s.

3. Three resistances of magnitude R each are connected in the form of an equilateral triangle of side a. The combination is placed in a magnetic field  $B = B_0 e^{-\lambda t}$  perpendicular to its plane. The induced current in the circuit is given by



- (A)  $\left( \frac{a^2 \lambda}{2\sqrt{3}R} B_0 \right) e^{-\lambda t}$   
 (B)  $\left( \frac{a^2 \lambda}{4\sqrt{3}R} B_0 \right) e^{-\lambda t}$   
 (C)  $\left( \frac{a^2 B_0}{\lambda 4\sqrt{3}R} \right) e^{-\lambda t}$   
 (D)  $\left( \frac{a^2 B_0 R}{\lambda 4\sqrt{3}} \right) e^{-\lambda t}$
4. Two convex lenses of focal length 10 cm and 20 cm respectively placed coaxially and are separated by some distance 'd'. The whole system behaves like a concave lens for rays coming from infinity. One of the possible value of 'd' is :
- (A) 40 cm  
 (B) 25 cm  
 (C) 15 cm  
 (D) 20 cm

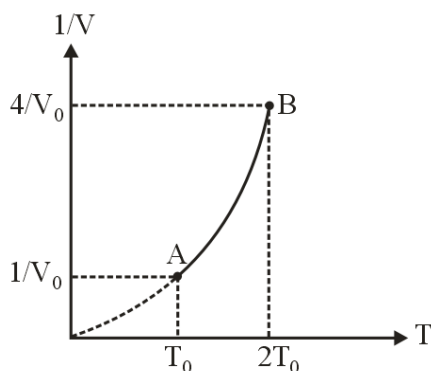
5. Four charges of  $8\mu\text{C}$ ,  $-6\mu\text{C}$ ,  $2\mu\text{C}$  and  $-4\mu\text{C}$  are placed at corners of a square placed in  $xy$  plane with its center at origin. The electric potential due to these charges is zero everywhere on the line

- (A)  $x = y, z = 0$  (B)  $x = 0 = z$   
(C)  $x = 0 = y$  (D)  $x = z, y = 0$

6. A particle starts oscillating simple harmonically from its extreme position at  $t = 0$ . The time period of oscillation is  $T$ . The ratio of kinetic energy and potential energy of particle at  $t = \frac{T}{6}$  is

- (A) 2 : 1 (B) 3 : 1  
(C) 4 : 1 (D) 1 : 4

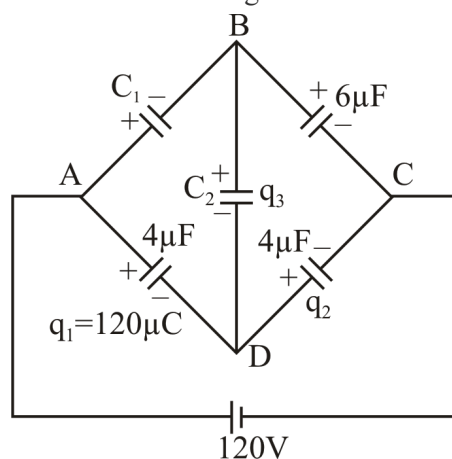
7. Figure shows a parabolic curve between  $\frac{1}{V}$  and  $T$  for a mixture of a gas undergoing adiabatic process.



The value of adiabatic exponent ( $\gamma$ ) of mixture is

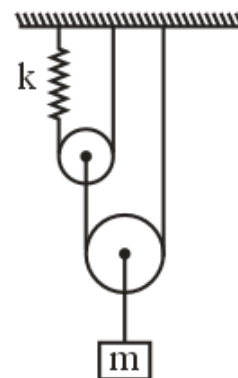
- (A)  $\frac{3}{2}$  (B)  $\frac{5}{3}$   
(C)  $\frac{7}{5}$  (D)  $\frac{5}{2}$

8. In the circuit shown in figure.



- (A)  $q_2 = 280\mu\text{C}$   
(B)  $q_2 = 360\mu\text{C}$   
(C)  $q_3 = 360\mu\text{C}$   
(D) It is not possible to find  $q_2$  and  $q_3$  unless  $C_1$  and  $C_2$  are known.

9. Block of mass  $m = 4\text{ kg}$  as shown in figure is slightly displaced and released. The spring constant  $k = 100\text{ N/m}$ . Pulleys are massless and frictionless. The time period of oscillation of block will be :



- (A)  $\frac{\pi}{2}\text{ s}$  (B)  $\frac{\pi}{4}\text{ s}$   
(C)  $\frac{\pi}{10}\text{ s}$  (D)  $\frac{2\pi}{3}\text{ s}$

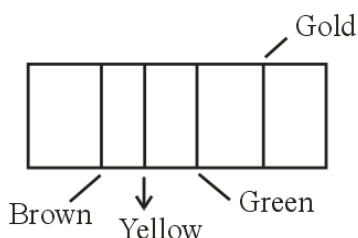
10. Two different perfect monoatomic gases at absolute temperature 300 K and 200 K having number of molecules  $2 \times 10^{18}$  and  $3 \times 10^{18}$  are mixed in a container of negligible heat capacity. Temperature of mixture is :

(A) 210 K (B) 240 K  
(C) 260 K (D) 280 K

11. 100 J of heat is given to a monatomic gas sample slowly such that pressure remains constant. The work done by gas is :

(A) 40 J (B) 80 J  
(C) 60 J (D) 20 J

12. A resistance is shown in the figure. Its value (in  $\Omega$ ) and tolerance are given respectively.



(A)  $14 \times 10^5$ , 5 % (B)  $14 \times 10^5$ , 10 %  
(C)  $27 \times 10^5$ , 10% (D)  $27 \times 10^5$ , 20%

13. Four simple harmonic motions

$$x_1 = 8 \sin 100 \pi t$$

$$x_2 = 6 \sin (100 \pi t + \frac{\pi}{2})$$

$$x_3 = 4 \sin (100 \pi t + \pi)$$

$$x_4 = 2 \sin (100 \pi t + \frac{3\pi}{2})$$

are superimposed on each other. The resulting amplitude of vibration is :

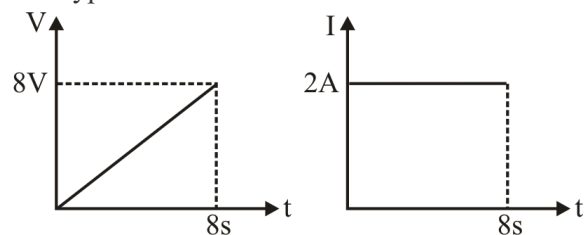
(A)  $4\sqrt{2}$  unit (B) 10 unit  
(C) 4 unit (D) 20 unit

14. A magnetic needle is free to rotate in vertical plane which makes an angle of  $60^\circ$  with magnetic meridian. If the needle stays in a direction making an angle of  $\tan^{-1} \left( \frac{2}{\sqrt{3}} \right)$

with horizontal. What would be the dip at that place.

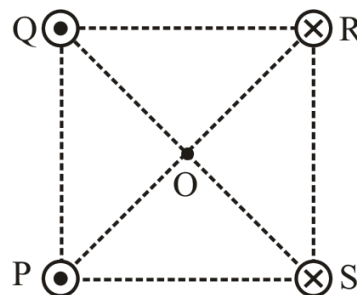
(A)  $37^\circ$  (B)  $30^\circ$   
(C)  $45^\circ$  (D)  $53^\circ$

15. Voltage versus time and current versus time graph of a circuit element is shown in figure. The type of circuit element is :



(A) resistance of  $2\Omega$  (B) capacitance of 1F  
(C) capacitance of 2F (D) none

16. Four long straight long wire are placed at corners of square PQRS. All the wires carry equal currents. Currents in wire P and Q are outwards and in R and S are inwards. The magnetic field at O is along :



(A) PQ (B) QR  
(C) SP (D) RS

17. One mole of an ideal monatomic gas undergoes the process  $P = 2\sqrt{T}$ . The molar specific heat of gas is

(A)  $\frac{3R}{4}$  (B)  $\frac{9R}{4}$   
(C)  $2R$  (D)  $\frac{2R}{3}$

18. A charged particle enters in a uniform magnetic field with velocity vector at an angle  $30^\circ$  with field. The pitch of helical path is  $p$ . The radius of helix will be :

(A)  $\frac{p}{2\pi}$  (B)  $\frac{p}{2\sqrt{3}\pi}$   
(C)  $\frac{2\pi p}{3}$  (D)  $\frac{p}{\sqrt{3}\pi}$

19. A heat pump operating between reservoirs at  $-5^\circ\text{C}$  and  $20^\circ\text{C}$  requires an electrical input of  $1.2\text{ kJ}$  in each cycle. The heat dissipated into hot reservoir in each cycle.

(A)  $12.8\text{ kJ}$  (B)  $11.7\text{ kJ}$   
(C)  $14\text{ kJ}$  (D)  $10\text{ kJ}$

20. A charged particle of mass ' $m$ ' and charge  $q$  is moving directly towards a fixed charge  $Q$ . When it is at a distance ' $d$ ' from fixed charge  $Q$  it has velocity  $v$ . The distance of closest approach will be. (Here,  $\frac{\pi\epsilon_0 mv^2}{qQ} = \frac{2}{d}$ )

(A)  $\frac{d}{2}$  (B)  $\frac{d}{3}$   
(C)  $\frac{d}{5}$  (D)  $\frac{3d}{4}$

## SECTION-II : (Maximum Marks: 20)

This section contains 10 questions. Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct integer value (in decimal notation, the answer should be rounded off to the nearest Integer).

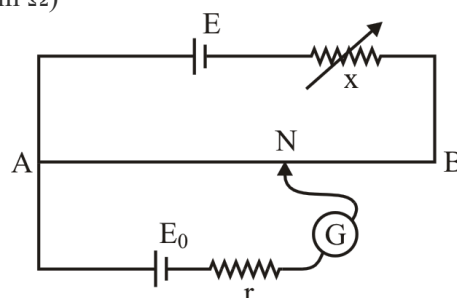
Answer to each question will be evaluated according to the following marking scheme:

**Full Marks** : +4 If correct answer is selected.

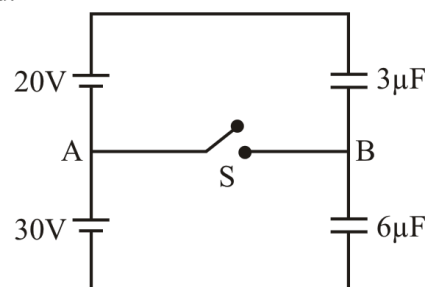
**Zero Marks** : 0 If none of the option is selected.

**Negative Marks** : -1 If wrong option is selected.

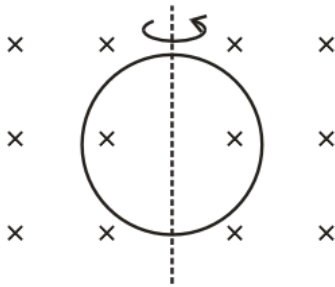
1. There is a potentiometer wire AB of resistance  $100\ \Omega$  and rheostat of variable resistance ' $x$ ' as shown in figure. For  $x = 0$ , null point is found at  $30\text{ cm}$  from A. For unknown value of  $x$  null point was  $40\text{ cm}$  from A. What is the value of  $3x$  (in  $\Omega$ )



2. What amount of charge (in  $\mu\text{C}$ ) will flow through switch S from A to B, when switch S is closed.

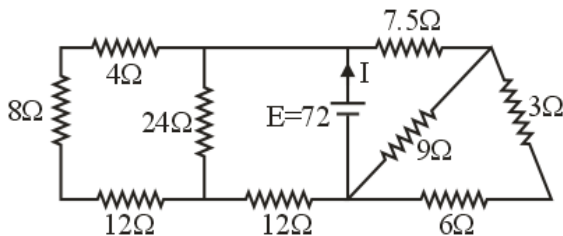


3. A particle of charge  $q$  and mass  $m$  starts moving from origin under action of an electric field  $\vec{E} = E_0 \hat{i}$  N/C and magnetic field  $\vec{B} = B_0 \hat{k}$  T. Its velocity at  $(x, 3, 0)$  is  $(4\hat{i} + 3\hat{j})$  m/s. The value of  $x$  (in m) is  $\frac{pm}{2qE_0}$ . The value of  $p$  is
4. A circular loop of area  $10 \text{ cm}^2$  and resistance  $4\Omega$  is rotated in a magnetic field  $B = 2\text{ T}$  through  $180^\circ$  in 20s as shown in figure.



If magnitude of amount of charge flown in circular loop in that interval is  $x \mu\text{C}$  then find value of  $x$ .

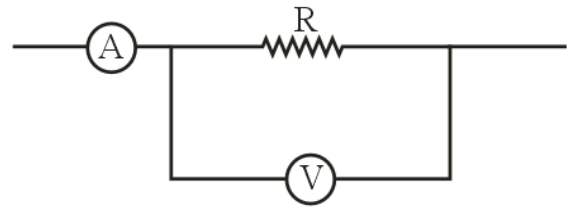
5. A circuit diagram is shown in figure



Find current  $I$  (in A) flowing through battery  $E$ .

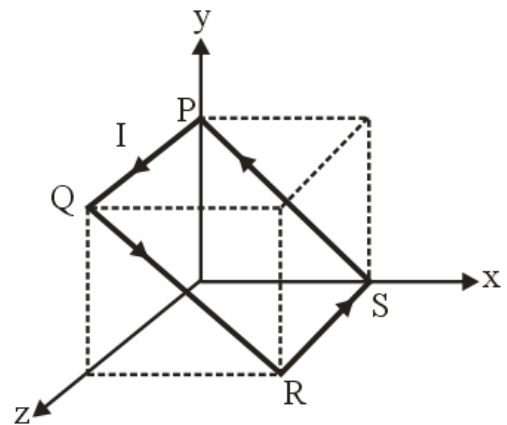
6. An astronomical telescope has angular magnification of magnitude 6 for distant objects. The separation between object and eye piece is 42 cm and final image is formed at infinity. Find focal length of objective (in cm).

7. The reading of ammeter is 10 A and voltmeter is 100 V as shown in figure.



If resistance of voltmeter is  $3000 \Omega$  then find value of resistance ' $R$ '.

8. The electric potential existing in space is given as  $V(x, y, z) = 10(xy + yz + zx)$  in SI unit. Find the value of electric field (in N/C) at point  $(2, 2, 2)$  (approximately).
9. An ice cube of mass 400 g at temperature  $0^\circ\text{C}$  melts to water at  $0^\circ\text{C}$ . Heat is supplied from the surrounding air at slightly above  $0^\circ\text{C}$ . The process takes place very slowly. Find the change in entropy (J/K) of ice when it has all melted. (Latent heat of ice =  $334 \text{ kJ/kg}$ )
10. A rectangular loop PQRS is formed by joining the corners of cube of side 20 cm as shown in figure. If  $I = 8 \text{ A}$  and  $\vec{B} = 0.4\hat{i} \text{ T}$  then the magnitude of torque on loop is  $P \times 10^{-3} \text{ N-m}$ . Find the value of  $P$ .



SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

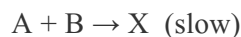
**Full Marks** : +4 If correct answer is selected.

**Zero Marks** : 0 If none of the option is selected.

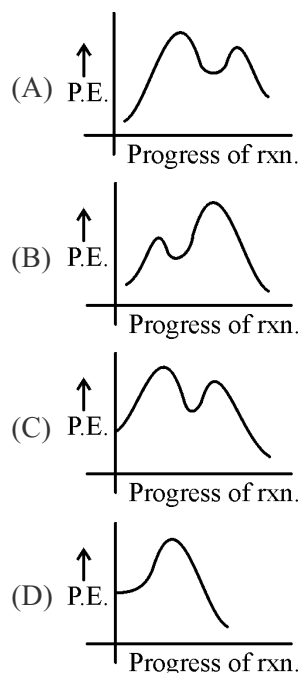
**Negative Marks** : -1 If wrong option is selected.

- Joseph Priestley discovered sweet smelling, colorless laughing gas, nitrous oxide ( $\text{N}_2\text{O}$ ), in 1763.  $\text{N}_2\text{O}$  has been widely used as an anesthetic for medical and dental applications. Nitrogen dioxide ( $\text{NO}_2$ ) is a toxic reddish brown gas that is the major component of smog. Nitrogen oxide ( $\text{NO}$ ) is a colorless, non-toxic gas that readily oxidizes to nitrogen dioxide in the presence of oxygen. Using the reactions :  
 $4\text{NO}(\text{g}) \rightarrow 2\text{N}_2\text{O}(\text{g}) + \text{O}_2(\text{g}) \Delta H^0 = -139.56 \text{ kJ}$   
 $2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{NO}_2(\text{g}) \Delta H^0 = -69.70 \text{ kJ}$   
 Calculate the standard enthalpy change when 2 moles of nitrous oxide combine with three moles of oxygen to produce four moles of nitrogen dioxide.  
 (A)  $-209.26 \text{ kJ}$  (B)  $-139.86 \text{ kJ}$   
 (C)  $-69.86 \text{ kJ}$  (D)  $+0.16 \text{ kJ}$
- Which of the following statements is correct :  
 (A) All covalent solids are good conductors of electricity  
 (B) In rhombohedral crystal system, only primitive unit cell is possible  
 (C) Ionic solids are ductile and brittle.  
 (D) In molecular solids, constituted particles are held together through covalent bond.

- For an exothermic reaction whose mechanism is given as :



The progress of the reaction can be best described by :

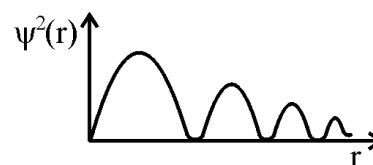


- From the following observations, predict the type of orbital.

Observation 1 :  $xy$ -plane acts as nodal plane.

Observation 2 : The angular function of the orbital intersect the three axis at origin only.

Observation 3 :  $R^2(r)$  v/s  $r$  curve obtained for the orbital is :

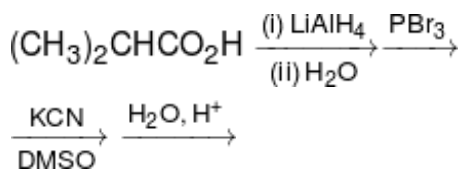


- (A)  $5p_z$  (B)  $6d_{xy}$   
 (C)  $6d_{x^2-y^2}$  (D)  $6d_{yz}$

5. When 5 mL of a 1.0 M HCl solution is mixed with 5 mL of a 0.1 M NaOH solution, temperature of solution increases by  $t^{\circ}\text{C}$ . Which of the following can be predicted from this observation :

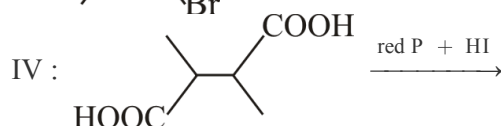
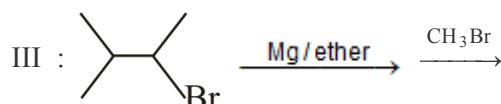
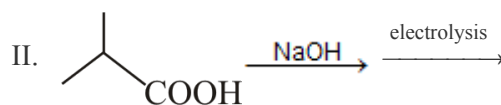
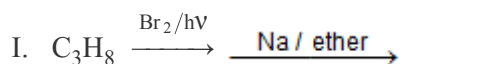
- (A) If 5 mL of 0.1M HCl is mixed with 5 mL 0.1 M  $\text{NH}_3$  solution, the temperature rise will be more than  $t^{\circ}\text{C}$
- (B) If 5 mL of 0.1 M  $\text{CH}_3\text{COOH}$  is mixed with 5 mL 0.1 M NaOH, the temperature rise will be less than  $t^{\circ}\text{C}$
- (C) If 10 mL of 1.0 M HCl mixed with 10 mL of 0.1 M NaOH, the temperature rise will be  $2t^{\circ}\text{C}$
- (D) If 10 mL of 0.1 M HCl is mixed with 10 mL of 0.1 M NaCl, the temperature rise will be  $t^{\circ}\text{C}$

6. Predict the major organic product of the following reactions.



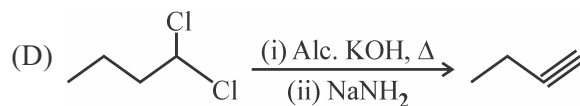
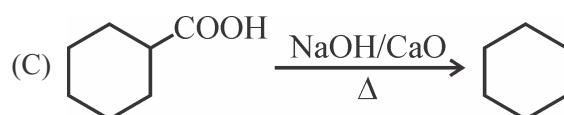
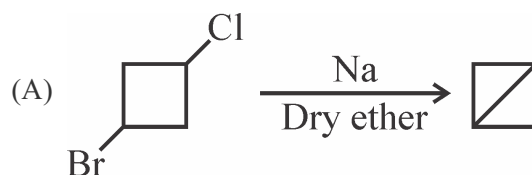
- (A)  $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_2\text{NH}_2$
- (B)  $(\text{CH}_3)_2\text{CHCHBrCO}_2\text{H}$
- (C)  $(\text{CH}_3)_2\text{C}=\text{CHCO}_2\text{H}$
- (D)  $(\text{CH}_3)_2\text{CHCH}_2\text{CO}_2\text{H}$

7. Which of the following reaction produces the same product :

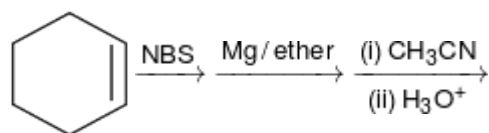


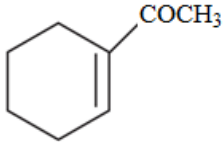
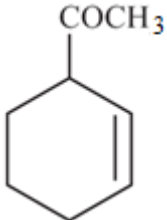
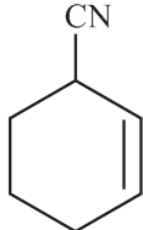
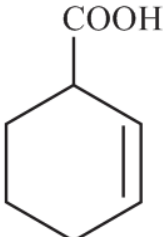
- (A) I, & II
- (B) I, II, III & IV
- (C) I, III & IV
- (D) II, III & IV

8. Which of the following is incorrectly matched with its major product :

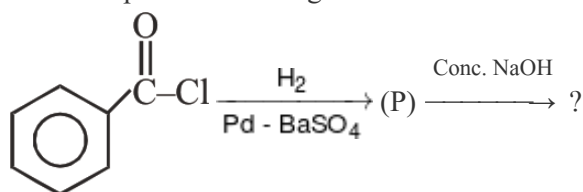


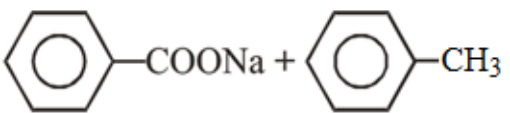
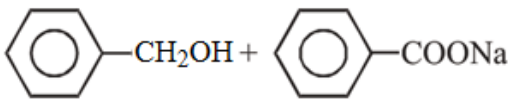
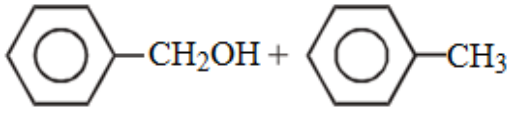
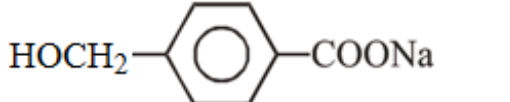
9. End product of the following sequence of reactions is :



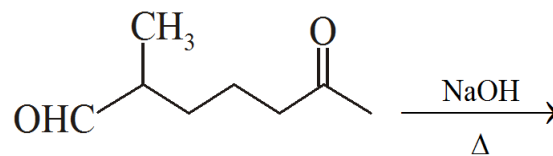
- (A)  (B) 
- (C)  (D) 

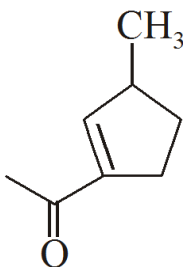
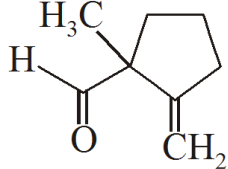
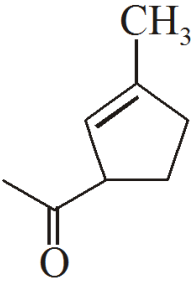
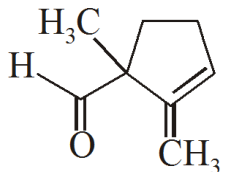
10. The final products of the given reaction is :

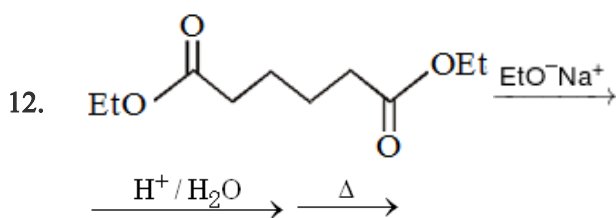


- (A)  (B) 
- (C)  (D) 

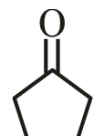
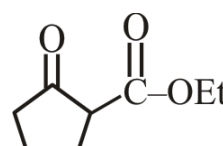
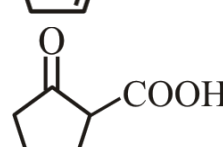
11. The major product obtained in the following reaction is :



- (A)  (B) 
- (C)  (D) 



Final product in above reaction sequence?

- (A) 
- (B) 
- (C) 
- (D) 

13.  $\text{XeF}_n + m\text{H}_2\text{O} \rightarrow \text{A (Contain Xe)} + \text{Other Products}$   
Regarding above reaction, which of the following option is correct :

- (A) When  $n = 6$  and  $m = 2$ , the hybridisation of Xe in A is  $\text{sp}^3\text{d}^2$ .
- (B) When  $n = 2$  and  $m = 1$ , then other products has a diatomic colored gas.
- (C) When  $n = 6$  and  $m = 3$ , then Electron pair structure of A is tetrahedral.
- (D) When  $n = 6$  and  $m = 1$ , the hybridisation of Xe in A is  $\text{sp}^3\text{d}$ .

14. Select the **CORRECT** option, for mentioned properties

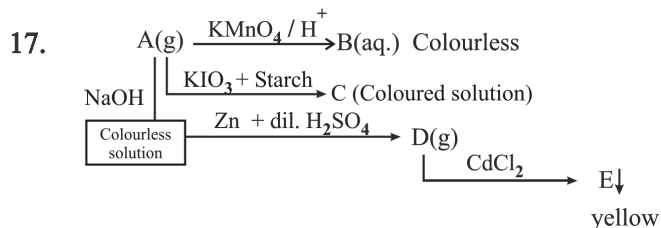
- (A)  $\text{Ba}^{2+} > \text{Ca}^{2+} > \text{Mg}^{2+} > \text{Be}^{2+}$  : (polarising power of cation)
- (B)  $\text{BeSO}_4 > \text{MgSO}_4 > \text{CaSO}_4 > \text{SrSO}_4$  : (thermal stability)
- (C)  $\text{BF}_3 < \text{BCl}_3 < \text{BBr}_3 < \text{BI}_3$  : (Lewis acidic nature)
- (D)  $\text{O}_2 < \text{KO}_2 < \text{K}_2\text{O}_2$  : (O-O bond order)

15. For  $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ , EAN of Fe will be :

- (A) 27 (B) 24  
(C) 35 (D) 29

16. Which of the following compounds produce  $\text{NH}_3$  on heating :

- (A)  $\text{NH}_4\text{Cl}$  (B)  $(\text{NH}_4)_2\text{S}$   
(C)  $(\text{NH}_4)_2\text{CO}_3$  (D) All of these



Gas 'A' is

- (A)  $\text{O}_3$  (B)  $\text{H}_2\text{S}$   
(C)  $\text{SO}_2$  (D)  $\text{NH}_3$

18. For a first order gaseous reaction half life is 25 minutes and reaction started with 1 atm pressure of pure A, then find final pressure of reaction mixture after 75 minutes.



- (A) 2 atm (B) 3 atm  
(C) 3.75 atm (D) 4.5 atm

19. Two mole of an ideal gas is isothermally compressed in two stages starting from (1 atm, 20 lit) to 10 lit. by constant external pressure  $P_1$  then upto 2 lit by constant external pressure  $P_2$ . Find work done in the process.

(A) 80 lit-atm (B) 100 lit-atm  
(C) 100J (D) 180 lit-atm

20.  $[\text{Ag}(\text{CN})_2]^-$  is formed :-

(A) In leaching of silver from  $\text{Ag}_2\text{S}$  using  $\text{NaCN}_{(\text{aq.})}$  &  $\text{O}_2$   
(B) In leaching of silver from native Ag using  $\text{NaCN}_{(\text{aq.})}$  &  $\text{O}_2$   
(C) When ppt. of  $\text{AgCl}$  reacts with excess  $\text{NaCN}$   
(D) All of the above

**SECTION-II : (Maximum Marks: 20)**

**This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.**

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct integer value (in decimal notation, the answer should be rounded off to the nearest Integer).

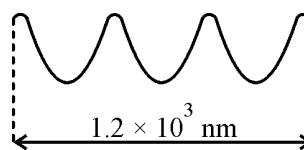
Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +4 If correct answer is selected.

*Zero Marks* : 0 If none of the option is selected.

*Negative Marks* : -1 If wrong option is selected.

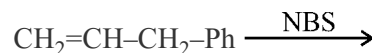
1. Consider the following wave representing electromagnetic radiation :



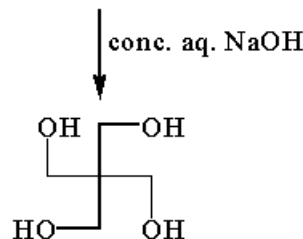
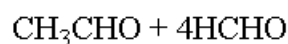
Calculate the energy of above wave (in eV) :

2. An ideal solution was prepared by dissolving some amount of cane sugar (non-volatile) in 0.9 moles of water. The solution was then cooled just below its freezing temperature ( $T_f = 271 \text{ K}$ ), where some ice get separated out. The remaining aqueous solution registered a vapour pressure of 700 torr at 373 K. Calculate the mass (in gm) of ice separated out ( $K_f$  of water =  $1.86 \text{ K.kg.mol}^{-1}$ )

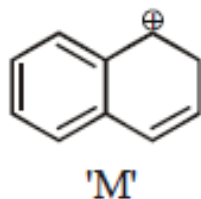
3. Total number of monobrominated product obtained in the given reaction is :



4. The number of Cannizaro reaction(s) that occurs in the given transformation is



5. How many resonating structures of carbocation 'M' as shown below are possible (excluding given resonating structure) :



6. Find the number of bidentate ligand from the abbreviated ligands given below.  
dien, bn,  $\text{ox}^{2-}$ , trien,  $\text{C}_2\text{O}_2\text{S}_2^{2-}$

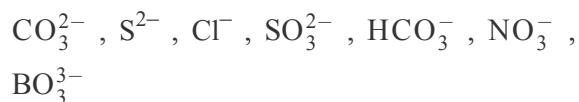
7. How much combinations from the following is/are correct :

|       | Metal | ore           | Chemical Formula                                  | Reduction process |
|-------|-------|---------------|---------------------------------------------------|-------------------|
| (i)   | Al    | Bauxite       | $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$ | Hall's process    |
| (ii)  | Cu    | copper pyrite | $\text{CuFeS}_2$                                  | Bessemerisation   |
| (iii) | Fe    | Haemetite     | $\text{Fe}_2\text{O}_3$                           | Smelting          |
| (iv)  | Pb    | Anglesite     | $\text{PbS}$                                      | Liquation         |

8. Find number of reagents which reduces  $\text{Cr}_2\text{O}_7^{2-}$  to  $\text{Cr}^{3+}$  in acidic aqueous medium.

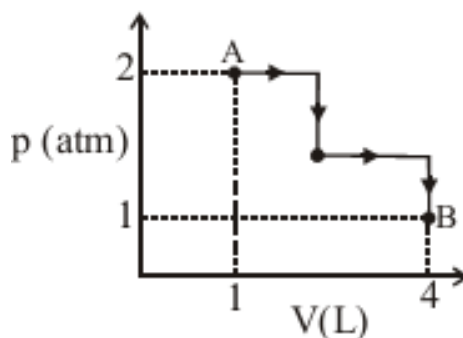


9. Find the number of acidic radical(s) which can form colourless gas when treated with dilute  $\text{H}_2\text{SO}_4$  ?



10. Calculate entropy change (in cal/K) for 1 mole of diatomic ideal gas undergoing for a process A to B as shown in diagram.

( $R = 2 \text{ cal/mole-K}$  ;  $\ln 2 = 0.7$ )



SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

**Full Marks** : +4 If correct answer is selected.

**Zero Marks** : 0 If none of the option is selected.

**Negative Marks** : -1 If wrong option is selected.

1. Consider  $f(x) = \begin{cases} x^2 - 2 & , x < 0 \\ x & , 0 \leq x < 2 \\ (x-2)^2 & , 2 \leq x < 4 \\ x-4 & , x \geq 4 \end{cases}$ .

If the equation  $f(x) = k$  has four roots, then  $k$  can be equal to :

- (A) -2 (B) 0  
(C) 2 (D) No value of  $k$

2. Let  $f(x) = \begin{cases} \frac{1}{|x-1|} & , -\infty < x < 0 \\ |x|^2 + 1 & , 0 \leq x < 1 \\ 2x & , 1 \leq x < 2 \\ \frac{x^2-1}{x-1} & , \text{otherwise} \end{cases}$ , then

$f(x)$  is discontinuous in  $(-\infty, \infty)$  at

- (A) two points (B) three points  
(C) one point (D) no point

3. If  $y = x + e^x$ , then at  $x = 1$ ,  $\frac{d^2y}{dx^2}$  is equal to

- (A)  $e$  (B)  $\frac{-e}{(1+e)^3}$   
(C)  $\frac{-e}{(1+e)}$  (D)  $\frac{-e}{(1+e)^2}$

4. If  $f'(x)$  is non-decreasing function for which  $f'(0) = 0$ ,  $f(0) = 5$  and  $f''(x) - f'(x) \leq 0$   $x \in \mathbb{R}$ , then the area bounded by the  $y = f(x)$  and the lines  $x = \frac{1}{2}$  and  $x = \frac{23}{2}$  does not lie in the interval :

- (A) [50, 55]  
(B) [50, 60]  
(C) [51, 54]  
(D) [40, 75]

5. Let  $f(x) = \begin{cases} 3x + 2\sin^{-1}x + p & , x \leq 0 \\ x^4 - x^3 + x^2 - qx + r & , x > 0 \end{cases}$ .

If  $f(x)$  is differentiable at  $x = 0$ , ( $p \neq 0$ ), then

- (A)  $(p-r)^2 + (q+5)^2 = 0$   
(B)  $(p+r)^2 + (q+5)^2 = 0$   
(C)  $(p-r)^2 + (q-5)^2 = 0$   
(D)  $(p+r)^2 + (q-5)^2 = 0$

6. The interval on which the function

$f(x) = \log_{1/3}(x^2 - 5x + 6)$  is monotonic increasing is

- (A)  $\left(2, \frac{5}{2}\right) \cup (3, \infty)$   
(B)  $(3, \infty)$   
(C)  $(-\infty, 2)$   
(D)  $(-\infty, 2) \cup \left(\frac{5}{2}, 3\right)$

7. Consider  $f(x) = \begin{cases} x(x+3) & , -1 \leq x < 0 \\ -\sin x & , 0 \leq x < \frac{\pi}{2} \\ -(1+\cos x) & , \frac{\pi}{2} \leq x \leq \pi \end{cases}$

The sum of maximum and minimum values of  $f(x)$  is

- (A) -2 (B) -1 (C) 0 (D) 2

8. The equation of normal to the curve  $x + y = x^y$ , where it cuts x-axis is

- (A)  $y = x + 1$   
(B)  $y = -x + 1$   
(C)  $y = x - 1$   
(D)  $y = -x - 1$

9.  $\int \frac{x^4 + x^3 - 1}{x^2 \sqrt{x^4 + 1 + 2x^3}} dx$  is equal to (where C is constant of integration)

- (A)  $\frac{\sqrt{x^4 + 2x^3 + 1}}{x^{1/2}} + C$   
(B)  $\frac{\sqrt{x^4 + 2x^3 + 1}}{x} + C$   
(C)  $\frac{\sqrt{x^4 + 2x^3 + 1}}{x^{3/2}} + C$   
(D)  $\frac{\sqrt{x^4 + 2x^3 + 1}}{x^3} + C$

10.  $\int_{\pi/4}^{\pi/2} [\sin x + |\cos x| + [\operatorname{cosec} x]] dx$  is equal to

(where  $[.]$  denotes greatest integer function)

- (A)  $\frac{\pi}{4}$  (B)  $\frac{\pi}{2}$   
(C)  $\pi$  (D)  $\frac{3\pi}{2}$

11. The value of  $\cos \left( \sin^{-1} \left( \tan \left( \cos^{-1} \left( \sin \left( \tan^{-1} \frac{4}{3} \right) \right) \right) \right) \right)$  is equal to :

- (A)  $\frac{3}{5}$  (B)  $\frac{3}{4}$  (C)  $\frac{\sqrt{7}}{4}$  (D)  $\frac{4}{5}$

12. Let  $f(x) = \max. \{|x^2 - 2|x||, |x|\}$ , then number of points where  $f(x)$  is non derivable, is :

- (A) 2 (B) 3 (C) 4 (D) 5

13. The tangent of acute angle between the curves  $y = |x^2 - 1|$  and  $y = \sqrt{7 - x^2}$  at their points of intersection is :

- (A)  $\frac{5\sqrt{3}}{2}$  (B)  $\frac{3\sqrt{5}}{2}$   
(C)  $\frac{5\sqrt{3}}{4}$  (D)  $\frac{3\sqrt{5}}{4}$

14. The area bounded by the curves  $f(x) = 9x^2 - 9x + 2$ ,  $g(x) = 9x^2 - 18x + 8$  and  $x = 1$  is :

- (A)  $\frac{1}{3}$  (B)  $\frac{1}{2}$  (C)  $\frac{2}{3}$  (D)  $\frac{3}{4}$

15.  $\int_{-\pi/4}^{\pi/4} \frac{(1 + \sin(x - \frac{\pi}{4})) \sqrt{2}}{\sqrt{2} - \cos |x| + \sin |x|} dx$  is equal to

- (A)  $\ln \left( \frac{\sqrt{2} - 1}{\sqrt{2}} \right) + \sqrt{2}$   
(B)  $\ln \left( \frac{\sqrt{2} - 1}{\sqrt{2}} \right) + \sqrt{2} + \frac{3\pi}{4}$   
(C)  $\ln \left( \frac{\sqrt{2}}{\sqrt{2} - 1} \right) + \sqrt{3} + \frac{3\pi}{4}$   
(D)  $\ln \left( \frac{\sqrt{2}}{\sqrt{2} - 1} \right) + \sqrt{3}$

16. If  $a \sin \theta + b \cos \theta = a \operatorname{cosec} \theta - b \sec \theta = 1$ , the value of  $a^2 + b^2 + b^{4/3} - b^{2/3} = k$  and

$$\beta + 2 \int_0^k x^2 e^{-x^2} dx = \int_0^k e^{-x^2} dx, \text{ then the}$$

value of  $\beta + k$  is equal to

(A)  $\frac{1+e}{e}$

(B)  $\frac{1-e}{e}$

(C) 0

(D)  $1+e$

17. If

$$\int \frac{x^6(x+1)}{\sqrt{5x^{10}+6x^9+x^4}} dx = \frac{\sqrt{5x^6+6x^5+1}}{k} + C$$

then  $k =$

(A) 10

(B) 15

(C) 5

(D) None

18.  $f'(x) = f(x) + \int_0^1 f(x) dx$  and  $f(0) = 1$ , then

$$\int f(x) dx \text{ is equal to } \underline{\hspace{2cm}}$$

(Where C is constant of integration)

(A)  $\frac{2}{3-e}e^x + \left(\frac{3-e}{1-e}\right)x + C$

(B)  $\frac{2}{3-e}e^x + \left(\frac{1-e}{3-e}\right)x + C$

(C)  $\frac{2}{2-e}e^x + \left(\frac{e}{e-2}\right)x + C$

(D)  $\left(\frac{3}{2-e}\right)e^x + \left(\frac{1-e}{2-e}\right)x + C$

19. Tangents and normals at points P and Q to the parabola  $y^2 = 4x$  intersect at point T and point R (9, 6) respectively. Then find the length of tangent drawn from (-1, -1) to the circle circumscribing the quadrilateral PTQR.

(A) 4 (B) 8 (C) 3 (D) 1

20. The differential equation of all those lines which are at a unit distance from origin is-

(A)  $\left(y - x \frac{dy}{dx}\right)^2 = 1 - \left(\frac{dy}{dx}\right)^2$

(B)  $\left(y + x \frac{dy}{dx}\right)^2 = 1 + \frac{dy}{dx}$

(C)  $\left(y - x \frac{dy}{dx}\right)^2 = 1 + \left(\frac{dy}{dx}\right)^2$

(D)  $\left(y + x \frac{dy}{dx}\right)^2 = 1 - \left(\frac{dy}{dx}\right)^2$

#### SECTION-II : (Maximum Marks: 20)

**This section contains 10 questions. Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.**

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct integer value (in decimal notation, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

**Full Marks** : +4 If correct answer is selected.

**Zero Marks** : 0 If none of the option is selected.

**Negative Marks** : -1 If wrong option is selected.

1. If the eccentricities of hyperbola  $xy = c^2$  and  $x^2 - y^2 = a^2$  are  $e_1$  and  $e_2$ , then the value of  $200(e_1^4 + e_2^4)$  must be \_\_\_\_\_

2. If (5, 12) and (24, 7) are the foci of an ellipse passing through the origin and if  $e$  is the eccentricity, then the value of  $\left[\frac{1}{e}\right]$ , where  $[\cdot]$  represents the greatest integer function is \_\_\_\_\_

3.  $\int \frac{x^4 + 1}{x^6 + 1} dx = A \tan^{-1} \left( x - \frac{1}{x} \right) + B \tan^{-1} (x^3) + C$   
then value of  $(A + B)$  is

4. Let  $y = g(x)$  be the image of  $f(x) = x + \sin x$  about the line  $x + y = 0$ . If the area bounded by  $y = g(x)$ ,  $x$ -axis,  $x = 0$  and  $x = 2\pi$  is  $A$ , then  $\frac{A}{\pi^2}$  is

5. If  $\int \frac{1}{(1 + \sqrt{x})^8} dx = \frac{-2}{a(1 + \sqrt{x})^a} + \frac{2}{b(1 + \sqrt{x})^b} + C$   
(where  $C$  is constant of integration), then the value of 'a' is equal to

6. The locus of the mid point of the chords of the ellipse  $49x^2 + 16y^2 = 784$ , the tangents at the ends of which intersect on the circle  $x^2 + y^2 = 100$  is  $\frac{(490x^2 + 160y^2)^2}{784(x^2 + y^2)} = k$  then  $k$  is equal to

7. If the tangent to the conic  $y - 4 = x^2$  at (3, 13) touches the circle  $x^2 + y^2 - 4x + 2y + \lambda = 0$  (for some fixed  $\lambda$ ) at a point (a, b), then  $|37a + 37b|$  is equal to

8. Let  $f(x)$  is an even function such that  $f(x) + f(x - 3) = x(x - 3) + 1$  then

$$2 \int_0^3 \frac{f(x)}{x^2 - 3x + 1} dx \text{ is}$$

9. If area bounded by the curve  $y = \sqrt{\sin[x] + [\sin x]}$ , (where  $[\cdot]$  G.I.F.) and lines  $x = 1$  and  $x = \frac{\pi}{2}$  and  $x$ -axis is equal to  $\sqrt{\sin 1} \left( \frac{\pi}{a} - b \right)$ , then  $a + b =$

10. The area bounded by the curves  $|x + y| \leq 1$ ,  $|y - x| \leq 1$  and  $2x^2 + 2y^2 \geq 1$  is  $\left| a - \frac{\pi}{b} \right|$  then  $a + b$  is equal to

## Space for Rough Work

